

PROJECT REPORT

Water Turbidity Measurement System

Using Arduino & Turbidity Sensor

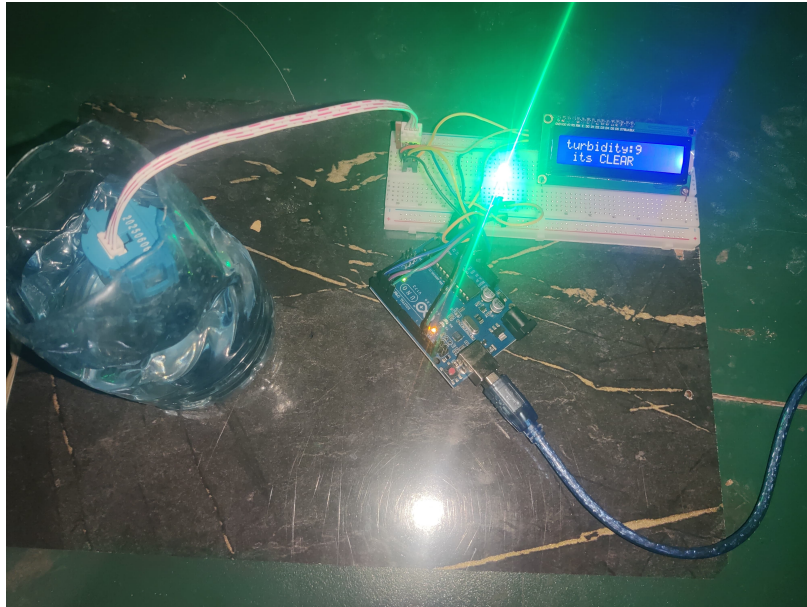


Fig. 1 – Prototype Testing: Clear Water (LCD reads CLEAR, Green LED ON)

Subject	Instrumentation and Measurements (MTE211)
Project Title	Water Turbidity Measurement System
Microcontroller	Arduino UNO R3
Sensor	Turbidity Sensor (Analog Output)
Display	16×2 I2C LCD Module
Indicator	RGB LED (Common Cathode)
Date	May 2026

1. Introduction

Water quality monitoring is a critical aspect of environmental engineering and public health. Turbidity — the cloudiness or haziness of a liquid caused by suspended particles — is one of the most fundamental indicators of water quality. High turbidity levels signal the presence of pollutants, sediments, or biological contaminants that can make water unsafe for drinking or industrial use.

This project presents a low-cost, real-time water turbidity measurement system built around an Arduino UNO microcontroller and an analog turbidity sensor. The system reads the optical sensor output, maps it to a turbidity percentage, displays the reading on a 16×2 I2C LCD, and provides an immediate visual alert through an RGB LED — green for clear water and red for dirty/turbid water.

2. Objectives

- To measure water turbidity in real time using an analog turbidity sensor.
- To display the turbidity reading and water quality status on a 16×2 I2C LCD.
- To provide a color-coded visual alert (Green = CLEAR, Red = DIRTY) via an RGB LED.
- To program the Arduino with a threshold-based decision algorithm.
- To demonstrate practical application of analog sensing and embedded systems in water quality monitoring.

3. Components & Materials

Component	Specification	Quantity	Purpose
Arduino UNO R3	ATmega328P, 5V	1	Main microcontroller
Turbidity Sensor	Analog output, 5V	1	Detect water cloudiness
16×2 I2C LCD	I2C addr: 0x27, 5V	1	Display turbidity value & status
RGB LED	Common cathode	1	Visual water quality indicator
Breadboard	Full-size	1	Circuit prototyping
Resistors	220Ω (for LED)	3	Current limiting for RGB pins
Jumper Wires	Male-to-male	~15	Electrical connections
Water Samples	Clear & turbid water	2	Testing
USB Cable	Type-B	1	Power & serial monitor

4. Circuit Diagram & Pin Connections

The circuit connects the turbidity sensor analog output to pin A0 of the Arduino. The I2C LCD is wired to the SDA and SCL pins. The RGB LED is connected to digital pins 2 (Green), 3 (Blue), and 4 (Red), each through a 220Ω current-limiting resistor.

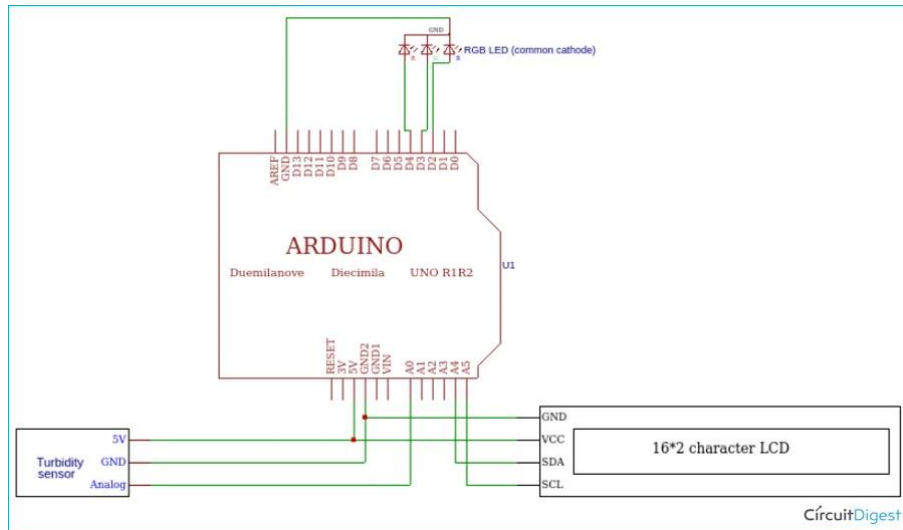


Fig. 2 – Circuit Diagram: Arduino + Turbidity Sensor + I2C LCD + RGB LED

Module	Module Pin	Arduino Pin	Notes
Turbidity Sensor	5V	5V	Power supply
Turbidity Sensor	GND	GND	Ground
Turbidity Sensor	Analog Out	A0	Reads 0–1023 ADC value
I2C LCD	VCC	5V	Power supply
I2C LCD	GND	GND	Ground
I2C LCD	SDA	A4 (SDA)	I2C Data
I2C LCD	SCL	A5 (SCL)	I2C Clock
RGB LED	R (Red)	D4	DIRTY indicator
RGB LED	G (Green)	D2	CLEAR indicator
RGB LED	B (Blue)	D3	Optional / off

5. Working Principle

The turbidity sensor operates on the principle of light scattering. An IR LED inside the sensor emits light through the water sample. Suspended particles scatter the light — the more particles present, the more light is scattered and the less reaches the phototransistor detector. This produces a lower analog voltage output for turbid water.

The Arduino reads this analog voltage via its 10-bit ADC (0–1023). The raw ADC value is mapped to a 0–100 turbidity scale using the `map()` function. A threshold of 25 is used: values below 25 indicate CLEAR water (green LED on), while values above 25 indicate DIRTY water (red LED on). Both the numeric reading and the status are displayed on the LCD in real time.

Step	Action	Output
1	IR LED emits light through water	Light passes / scatters based on particles
2	Phototransistor detects light	Analog voltage → Arduino A0
3	Arduino reads ADC (0–1023)	Raw sensor value printed to Serial
4	<code>map()</code> scales to 0–100	Turbidity percentage calculated
5	Threshold check (< or > 25)	Status = CLEAR or DIRTY
6	LCD displays value + status	"turbidity:X" + "its CLEAR/DIRTY"
7	RGB LED activates	Green = CLEAR Red = DIRTY

6. Arduino Code

The complete Arduino sketch is shown below in two parts — the setup/loop header and the threshold decision logic.

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

int sensorPin = A0;

void setup()
{
  Serial.begin(9600);

  lcd.init();
  lcd.backlight();

  pinMode(2, OUTPUT); // Green
  pinMode(3, OUTPUT); // Blue
  pinMode(4, OUTPUT); // Red
}

void loop()
{
  int sensorValue = analogRead(sensorPin);
  Serial.println(sensorValue);

  int turbidity = map(sensorValue, 0, 750, 100, 0);

  delay(100);
```

```
    lcd.setCursor(0, 0);
    lcd.print("turbidity:");
    lcd.print(" ");
    lcd.setCursor(10, 0);
    lcd.print(turbidity);

    delay(100);

    if (turbidity < 25) {
      digitalWrite(2, HIGH);
      digitalWrite(3, LOW);
      digitalWrite(4, LOW);

      lcd.setCursor(0, 1);
      lcd.print(" its CLEAR ");
    }

    if (turbidity > 25) {
      digitalWrite(2, LOW);
      digitalWrite(3, LOW);
      digitalWrite(4, HIGH);

      lcd.setCursor(0, 1);
      lcd.print(" its DIRTY ");
    }
  }
```

Fig. 3 (Left) – Setup & Loop | Fig. 4 (Right) – Threshold Logic & LCD Output

Code Explanation:

Function / Line	Purpose
#include <LiquidCrystal_I2C.h>	Imports I2C LCD library
LiquidCrystal_I2C lcd(0x27,16,2)	Initialises LCD at I2C address 0x27, 16 cols × 2 rows
analogRead(A0)	Reads raw sensor ADC value (0–1023)
map(val, 0, 750, 100, 0)	Converts ADC to turbidity % (inverted: low voltage = high turbid.
turbidity < 25	CLEAR threshold – green LED on, LCD shows 'its CLEAR'
turbidity > 25	DIRTY threshold – red LED on, LCD shows 'its DIRTY'
delay(100)	100ms refresh rate for stable readings

7. Testing & Results

The system was tested with two distinct water samples — clean tap water and dirty/turbid water (with added particulate matter). Results were observed on the LCD and confirmed by the RGB LED color change.

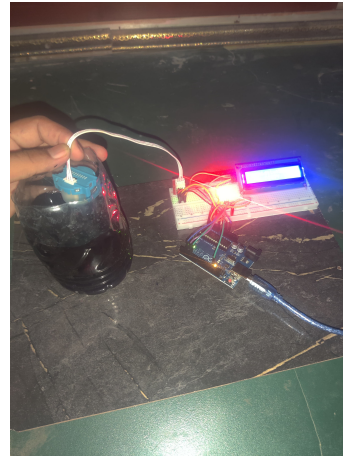
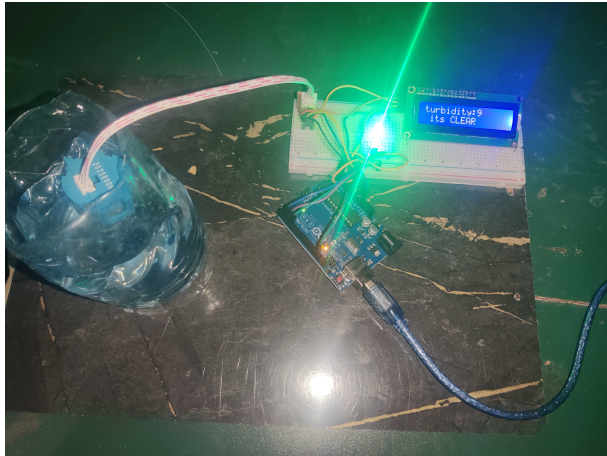


Fig. 5 – Clear Water: turbidity=9, LCD: CLEAR, Green LED ON Fig. 6 – Dirty Water: high turbidity, LCD: DIRTY, Red LED ON

Test	Water Sample	ADC Value	Turbidity %	LED Color	LCD Status
1	Clean tap water	~680	< 25	GREEN	its CLEAR
2	Slightly muddy	~500	~33	RED	its DIRTY
3	Highly turbid	~150	~80	RED	its DIRTY

8. Reference Setup

The image below shows the standard reference circuit configuration for interfacing a turbidity sensor with Arduino, as used in the design and verification of this project.

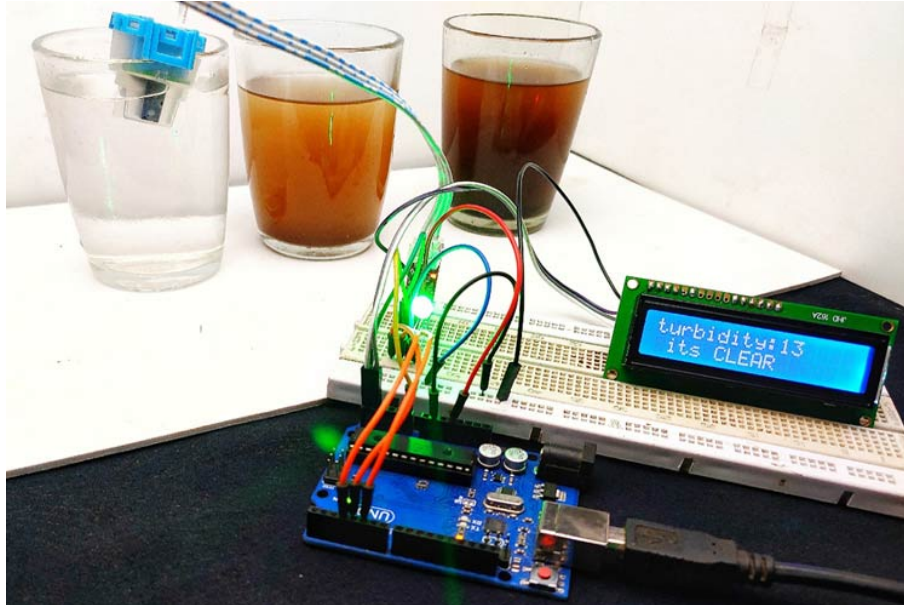


Fig. 7 – Reference Setup: Turbidity sensor in clear vs. turbid water samples

9. Advantages & Applications

Advantages

- Low cost — uses widely available Arduino and analog sensor.
- Real-time monitoring with 100ms refresh rate.
- Dual output — LCD display + RGB visual alert.
- Easily adjustable threshold in software.
- Compact and portable prototype design.

Applications

- Drinking water quality monitoring.
- Industrial wastewater discharge monitoring.
- Aquaculture / fish tank water control.
- River & lake pollution early-warning.
- Educational embedded systems demo.

10. Observations

- The LCD correctly displayed the turbidity value and status ('CLEAR' or 'DIRTY') for all test samples.
- The RGB LED changed color instantaneously upon crossing the turbidity threshold of 25.
- Clean tap water consistently produced ADC readings above 600 (turbidity < 25).
- Adding particulate matter (soil/mud) to water caused rapid ADC drop and triggered the DIRTY alert.
- The map() function provided a consistent 0–100 scale, making readings intuitive.
- Sensor readings stabilized within 1–2 seconds when immersed in a new water sample.
- The 100ms delay provided smooth, flicker-free LCD updates without noticeable lag.
- The I2C LCD significantly reduced wiring complexity compared to a parallel LCD connection.

11. Conclusion

The Water Turbidity Measurement System was successfully designed, built, and tested. The project demonstrated that an Arduino UNO combined with a low-cost analog turbidity sensor can reliably distinguish between clean and contaminated water in real time, with clear visual feedback via an RGB LED and a 16×2 I2C LCD display.

The threshold-based algorithm (turbidity < 25 → CLEAR, turbidity > 25 → DIRTY) proved effective across multiple test samples. The system deepened practical understanding of analog sensor interfacing, ADC value mapping, I2C communication, and embedded decision logic — all fundamental skills in electronics and embedded systems engineering.

Future improvements could include: adding a buzzer alarm for high turbidity, IoT integration for remote monitoring via Wi-Fi, data logging to an SD card, and a calibrated multi-level turbidity scale (e.g. CLEAN / MODERATE / HIGHLY POLLUTED) for more nuanced classification.